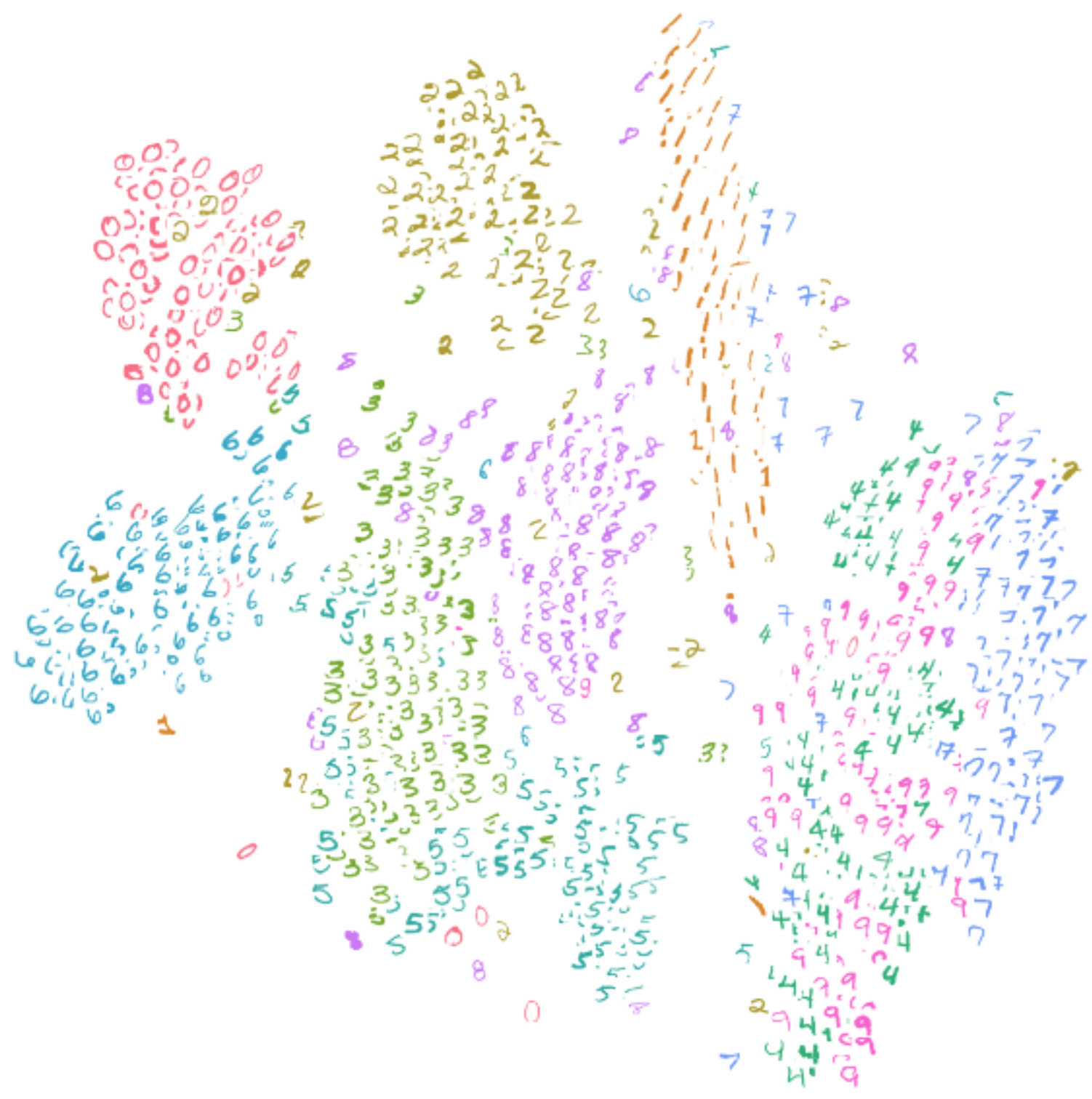


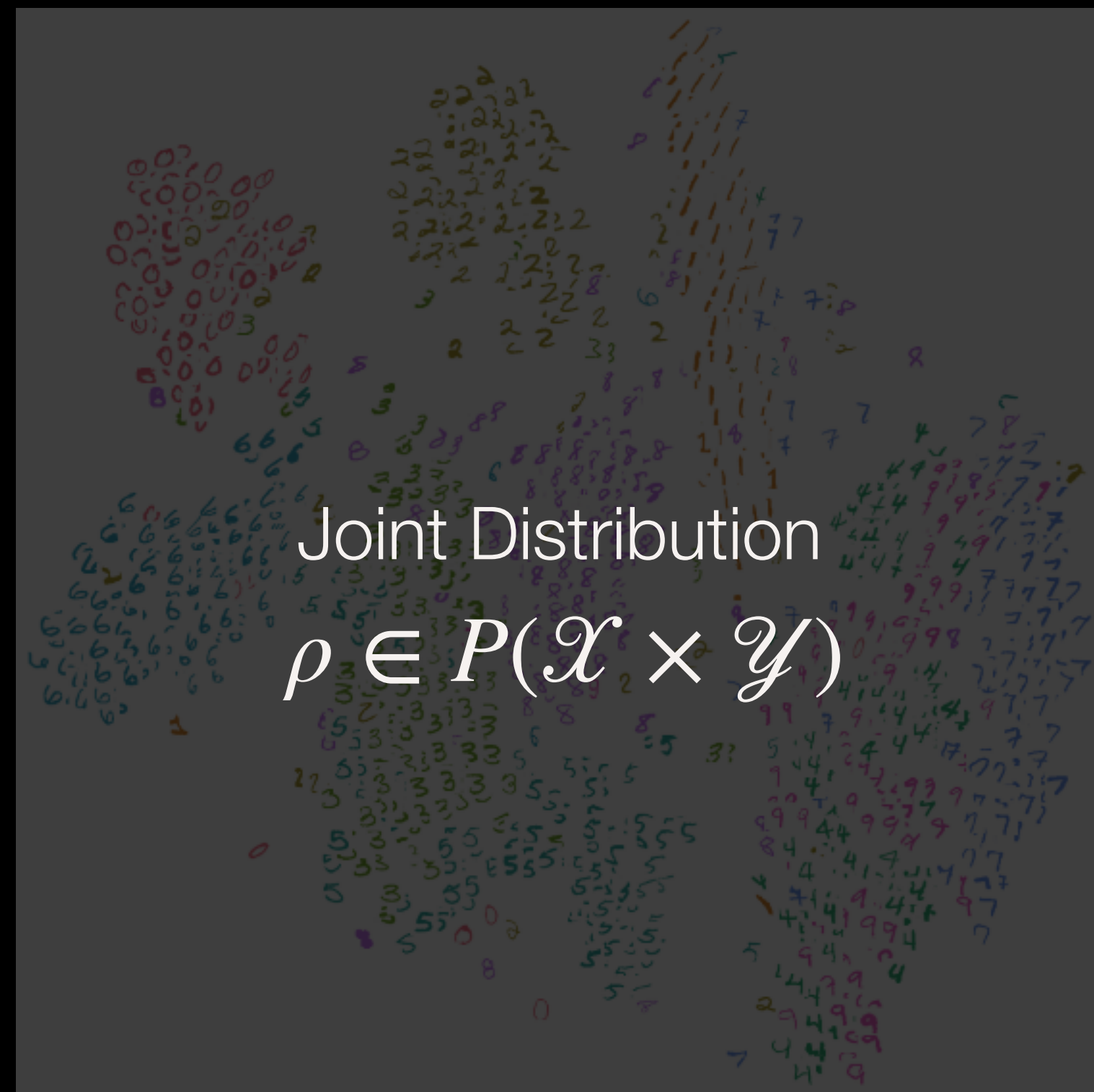




maximize similarity to another dataset

A-M & Fusi. *Dataset Dynamics via Gradient Flows in Probability Space*, ICMML 2021

Many goals, One Framework



Joint Distribution

$$\rho \in P(\mathcal{X} \times \mathcal{Y})$$

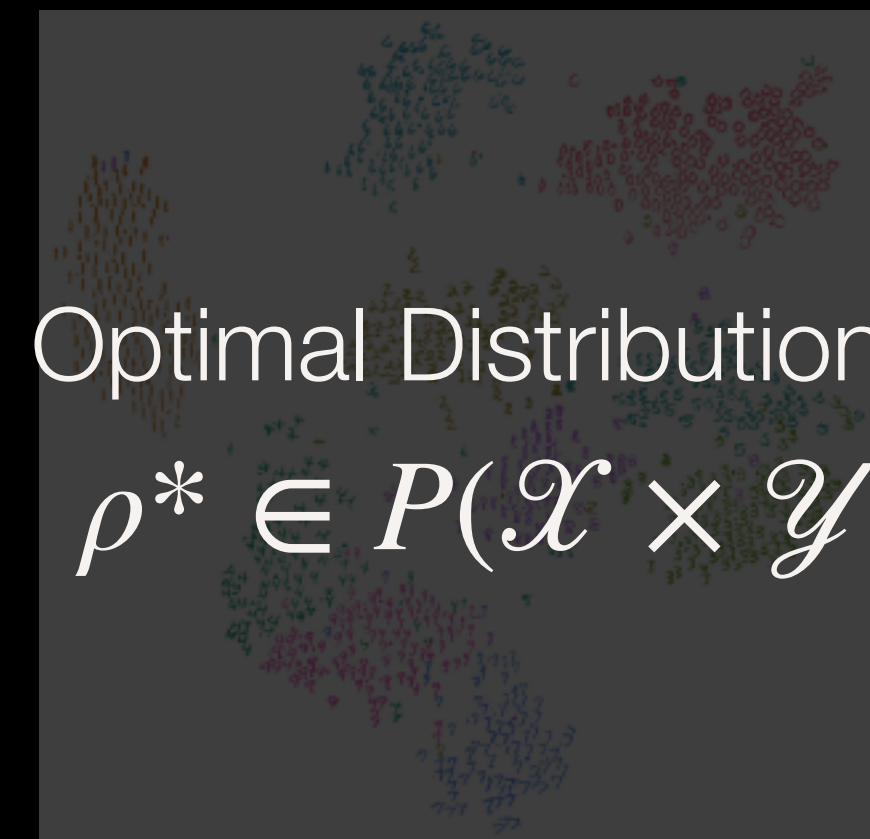
Initial Dataset

increase class separation

$$\min_{\rho} F(\rho)$$

hide protected/sensitive attributes

Optimization Objective



Optimal Distribution

$$\rho^* \in P(\mathcal{X} \times \mathcal{Y})$$

Optimization Solution

Dataset Optimization: Gradient Flows

Need to solve optimization problem over (**infinite dimensional**, **non-euclidean**!) probability space